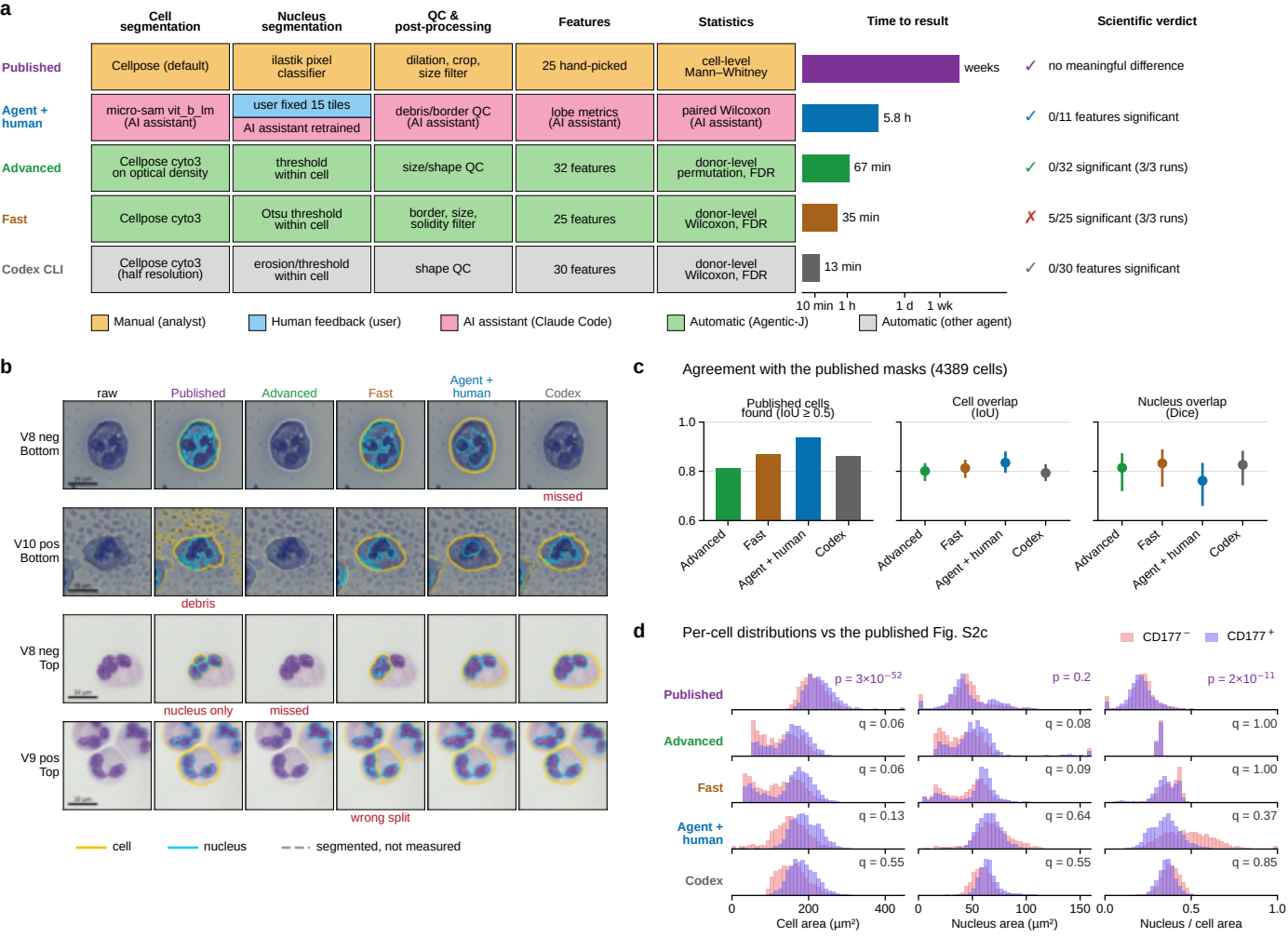

a
Cell segmentation
Nucleus segmentation
QC & post-processing
Features
Statistics
Time to result
Scientific verdict
Published
Cellpose (default)
ilastik pixel classifier
dilation, crop, size filter
25 hand-picked
cell-level Mann–Whitney
weeks
no meaningful difference
Agent + human
micro-sam vit_b_lm (AI assistant)
user fixed 15 tiles
AI assistant retrained
debris/border QC (AI assistant)
lobe metrics (AI assistant)
paired Wilcoxon (AI assistant)
5.8 h
0/11 features significant
Advanced
Cellpose cyto3 on optical density
threshold within cell
size/shape QC
32 features
donor-level permutation, FDR
67 min
0/32 significant (3/3 runs)
Fast
Cellpose cyto3
Otsu threshold within cell
border, size, solidity filter
25 features
donor-level Wilcoxon, FDR
35 min
5/25 significant (3/3 runs)
Codex CLI
Cellpose cyto3 (half resolution)
erosion/threshold within cell
shape QC
30 features
donor-level Wilcoxon, FDR
13 min
0/30 features significant
10 min 1 h 1 d 1 wk
Manual (analyst)
Human feedback (user)
AI assistant (Claude Code)
Automatic (Agentic-J)
Automatic (other agent)
b
raw
Published
Advanced
Fast
Agent + human
Codex
V8 neg Bottom
missed
V10 pos Bottom
debris
V8 neg Top
nucleus only
missed
V9 pos Top
wrong split
cell
nucleus
segmented, not measured
c
Agreement with the published masks (4389 cells)
Published cells found (IoU ≥ 0.5)
Cell overlap (IoU)
Nucleus overlap (Dice)
1.0
0.8
0.6
Advanced
Fast
Agent + human
Codex
d
Per-cell distributions vs the published Fig. S2c
CD177⁻
CD177⁺
Published
Advanced
Fast
Agent + human
Codex
p = 3×10⁻⁵²
p = 0.2
p = 2×10⁻¹¹
q = 0.06
q = 0.08
q = 1.00
q = 0.06
q = 0.09
q = 1.00
q = 0.13
q = 0.64
q = 0.37
q = 0.55
q = 0.55
q = 0.85
0
200
400
Cell area (μm²)
0
50
100
150
Nucleus area (μm²)
0.0
0.5
1.0
Nucleus / cell area